

```

In [10]: np.zeros((2,2))
Out[10]: array([[0., 0.],
               [0., 0.]])

In [11]: np.eye(3)
Out[11]: array([[1., 0., 0.],
               [0., 1., 0.],
               [0., 0., 1.]])

In [18]: import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
with tf.Session() as sess:
    fig, ax=plt.subplots()
    ax.plot(tf.random_normal([100]).eval(),
            tf.random_normal([100]).eval(),'o')
    ax.set_title('Sample random plot for Tensorflow')
    plt.savefig("result.png")

In [ ]:

```

Figure 5: Synthetic data set

the two variables as constants. It passes four numbers to the `constant()` function in Jupyter Notebook, resulting in an output.

Linear regression is the simple form of learning the parameters for any one-dimensional line. Let the data points be 'x' as one-dimensional and the class label 'y' generated by the linear equation:

$$y = mx + c$$

Here, 'm' and 'c' will be the learnable parameters that are calculated with the input. Now learn the parameters with TensorFlow by creating the artificial data set. To make this a bit hard, add a little bit of Gaussian noise to the data set, as follows:

$$y = mx + c + N(0, \epsilon)$$

Introduction to classification

The creation of a machine learning model that is used to assign discrete labels to its particular input is called classification. Every discrete value is called a class and handles supervised learning algorithms with a discrete

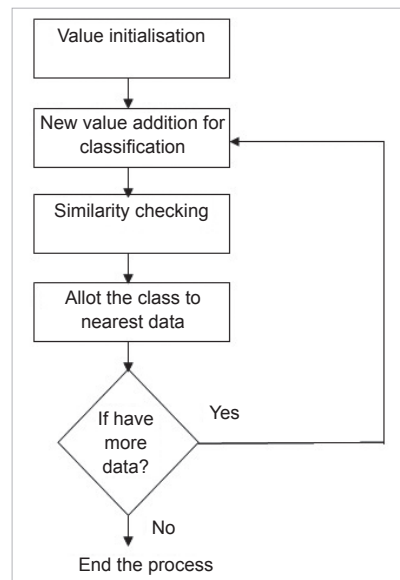


Figure 6: Flow chart for K-means clustering

output. Here, the input is typically a feature vector and the output is a class. If there are only two class labels in the input (Yes/No, True/False, On/Off), it

is called a binary classifier; otherwise, it is called a multiclass classifier.

Clustering data

Traversing files in TensorFlow:

Unsupervised learning basically contains patterns of data sets. Clustering is mainly an unlabelled data set and helps to understand the groups of data with feature vectors. The data set is divided into a number of segments, where each and every point can be treated as a centroid, i.e., points belong to the particular group or cluster. Figure 5 shows the synthetic data set tensor code.

K-Means clustering on synthetic data sets: The flowchart for K-means clustering is given in Figure 6.

Machine learning empowers us to develop complex applications with extraordinary precision. TensorFlow helps to develop these applications. **END** 🐧

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